PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

AI DRIVEN SMART HEALTH ADVISOR

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"AI DRIVEN SMART HEALTH ADVISOR"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: ……………..

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

The AI-Driven Smart Health Advisor is proposed to address the growing challenges of managing chronic diseases and responding to epidemic situations, particularly in resource-limited and underserved regions. Traditional healthcare systems often focus on reactive treatment rather than preventive care, which can lead to delayed intervention and increased health risks. This system aims to shift the paradigm toward preventive and proactive healthcare by leveraging Artificial Intelligence (AI) and data-driven decision-making.

The platform is designed using a lightweight and modular Python-based architecture, ensuring scalability and usability even in low-connectivity environments. It consists of two primary engines: a predictive engine and a guideline retrieval engine. The predictive engine analyzes user vitals such as blood pressure, fasting glucose levels, and Body Mass Index (BMI) using rule-based mathematical logic to classify health risks into different categories. The guideline retrieval engine utilizes Natural Language Processing (NLP) techniques to map user queries and symptoms to verified medical protocols and recommendations from trusted organizations such as WHO and NCDC.

In addition to risk assessment, the system incorporates multiple integrated functionalities, including symptom-based urgency detection, outbreak awareness using real-time public health data, and location-based healthcare facility recommendations. These features enable the system to provide personalized, real-time health advice while also considering the broader public health context.

The advisor is capable of identifying high-risk conditions, such as abnormal vital readings or symptoms indicative of infectious diseases like COVID-19 or Nipah virus, and generates appropriate alerts. It guides users through a structured decision-making process, helping them determine whether a situation can be managed at home, requires medical consultation, or demands immediate emergency attention.

Experimental evaluation demonstrates that the system consistently identifies critical health scenarios and delivers reliable recommendations. By integrating automated risk analysis with authoritative health knowledge, the proposed system reduces misinformation and enhances the reliability of health guidance.

Overall, the AI-Driven Smart Health Advisor contributes to improving health literacy, optimizing resource utilization, and empowering individuals to make informed healthcare decisions. It serves as a practical and accessible solution for bridging gaps in healthcare accessibility, especially in rural and underserved communities, and has the potential to support both long-term chronic disease management and immediate emergency response on a unified platform.

1. INTRODUCTION

1.1. Background and Context

The past ten years have seen an unprecedented increase in the prevalence rates of chronic non-communicable diseases and outbreaks of infectious diseases across the globe. Alongside the ongoing monitoring of hypertension and diabetes, emergent threats like Nipah and COVID-19 demand immediate access to verified, updated clinical guidelines. This burden is particularly compounded in remote or underserved areas, where access to physicians, diagnostic centers, and dependable health information is scarce, leaving populations to rely on incomplete or unproven information. Moreover, rapid urbanization, lifestyle changes, and lack of continuous monitoring tools have further accelerated the incidence of such diseases, making early detection and preventive care increasingly critical.

1.2. Problem Statement

Conventional healthcare delivery is highly reactive and episodic; people usually engage with the health system only once symptoms become critical. This delays risk identification, increases unnecessary hospitalizations, and burdens medical facilities. During epidemics, the rapid spread of misinformation on social media causes individuals to neglect crucial precautions or overwhelm hospitals with cases that could be managed at home. Existing AI applications are fragmented: symptom checkers rarely incorporate live vitals or authoritative regulatory guidance, and many IoMT predictive systems require heavy cloud infrastructure and constant broadband, making them impractical for rural settings. Additionally, the lack of integration between different health systems limits the effectiveness of current digital healthcare solutions.

1.3. Proposed Solution

To bridge these gaps, this project proposes an AI-driven Smart Health Advisor utilizing a dual-engine architecture. It transforms healthcare from reactive intervention to proactive preventive care without requiring advanced infrastructure. By integrating an explainable predictive engine for chronic risk assessment and a regulatory NLP engine for outbreak triage, the system delivers timely, trusted, and context-sensitive recommendations directly to personal devices, mitigating the spread of medical misinformation and preventing chronic health crises. Furthermore, the system is designed to be lightweight, scalable, and adaptable, enabling its deployment in low-resource environments while maintaining efficiency and reliability.

2. Methodology

2.1. System Architecture Overview

The system employs a service-oriented, three-layer architecture: a user interface, orchestration, and two inference engines. The orchestration layer directs numerical vital data to the predictive engine and free-text queries to the regulatory engine. The overall design is lightweight, modular, and robust against intermittent connectivity, allowing it to function efficiently on the simple hardware typically available in underserved areas.

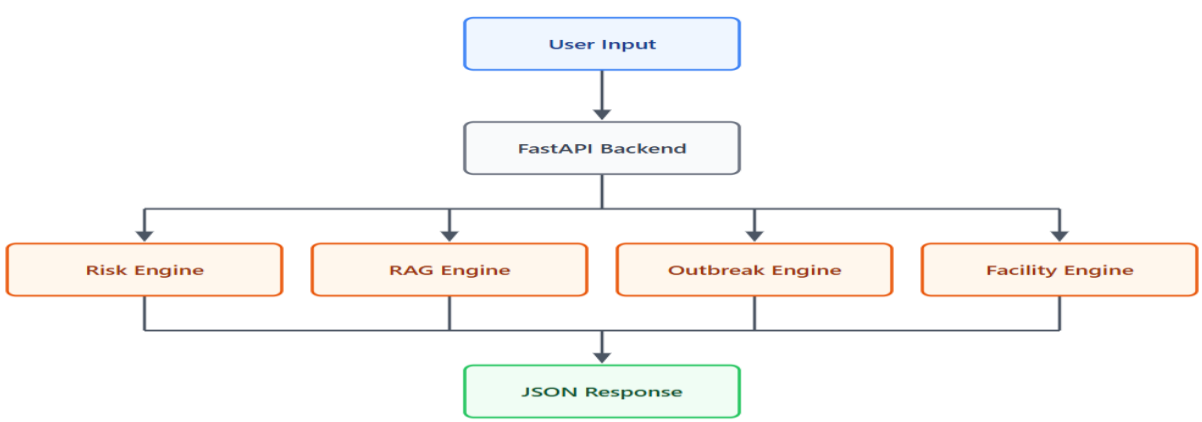
2.2. Predictive Health Engine

The Predictive Health Engine focuses on long-term preventive care by processing user-entered blood pressure and fasting glucose data. Instead of relying purely on black-box neural networks, it utilizes deterministic clinical thresholds based on standard hypertension and diabetes guidelines. This categorizes readings into normal, pre-risk, and high-risk bands. Once safety margins are breached, the engine triggers high-risk flags and provides actionable recommendations, including lifestyle changes and professional assessment schedules. The engine includes a plug-in layer utilizing Scikit-learn, allowing for the future integration of explainable Machine Learning models like Decision Trees or Random Forests.

2.3. Regulatory Inquiry Engine

The Regulatory Inquiry Engine manages outbreak-aware triage and misinformation control. When users input natural language descriptions of symptoms, the engine conducts a first-step tokenizing/pattern match to quickly identify high-priority conditions (e.g., severe fever, Nipah-related symptoms). For complex queries, it initiates a Retrieval-Augmented Generation (RAG) workflow. This engine queries a vectorized database of official WHO and NCDC protocols, condensing retrieved passages into brief, actionable instructions such as isolation requirements or environmental precautions, ensuring all advice is directly tied to authoritative regulations.

Figure 1: Methodology Architecture



3. Tools and Technologies

3.1. Frontend and Orchestration

- **React.js:** Used to develop a responsive and lightweight user interface that enables users to input health parameters such as blood pressure, glucose levels, BMI, and symptom descriptions. The interface is designed to be simple, intuitive, and optimized for low-end devices, ensuring accessibility in resource-constrained environments.
- **Node.js & Express.js:** Serve as the central orchestration layer, handling API routing, request processing, and communication between the frontend and backend services. This layer ensures efficient data flow by directing numerical inputs and natural language queries to the appropriate processing engines while maintaining scalability and performance.

3.2. Backend Inference Engines

- **Python & FastAPI:** Power the core predictive health engine by providing a high-performance REST API. FastAPI enables efficient validation and processing of incoming data, while Python implements rule-based and logical computations for assessing health risks based on vitals such as blood pressure, glucose, and BMI.
- **Node.js & LangChain.js:** Facilitate the Retrieval-Augmented Generation (RAG) pipeline for the regulatory inquiry engine. This component processes user queries using Natural Language Processing (NLP) techniques and retrieves relevant health guidelines from authoritative sources, ensuring context-aware responses.
- **MongoDB Atlas Vector Search:** Functions as a vector database to store and retrieve structured health guidelines and protocols (e.g., WHO, NCDC). It enables fast semantic similarity search, improving the accuracy and relevance of retrieved information.
- **Scikit-learn:** Provides a machine learning layer for the predictive engine, supporting interpretable models such as Decision Trees and Random Forests. These models enhance risk classification while maintaining transparency and explainability in predictions.
- **Internet of Medical Things (IoMT):** Represents the conceptual integration layer for future expansion, allowing continuous health monitoring through wearable devices. This enables real-time data collection and enhances the system's capability for proactive healthcare analysis and timely intervention.

4. Implementation

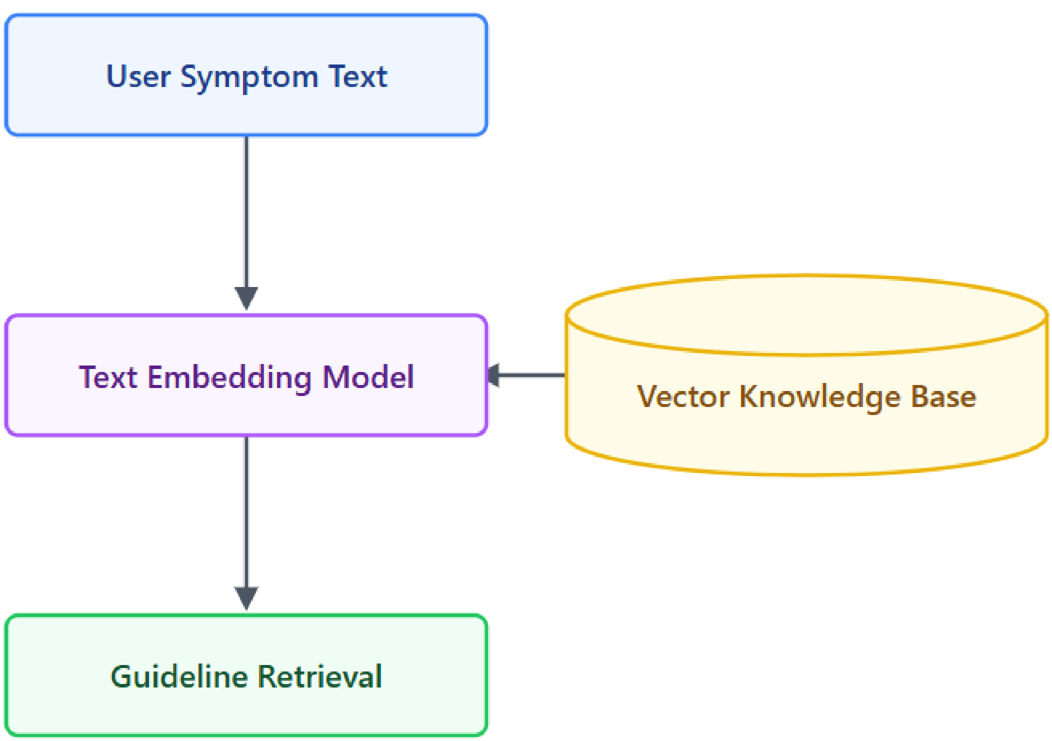
4.1. Data Routing and Processing

The implementation relies on decoupled microservices to ensure scale independence and deployment flexibility. When a user submits data via the React interface, the Express.js server evaluates the payload type. Numerical vitals are routed to the Python FastAPI microservice, where deterministic mathematical rules instantly assess chronic risk without heavy computational overhead.

4.2. RAG Pipeline Execution

Simultaneously, natural language queries are routed to the Node.js backend. LangChain.js embeds the user's query and performs a similarity search against the MongoDB Atlas Vector database. The system retrieves specific WHO/NCDC guidelines and synthesizes a context-aware response, encoding critical outbreak instructions (e.g., 21-day tracking periods for close contacts) directly into the system logic. Both engines return their assessments to the Express.js server, which compiles a unified health advisory displayed to the user.

Figure 2 – Rag Pipeline Implementation Model



5. Major Findings

5.1. Performance and Accuracy

- **Predictive Engine Accuracy:** Experimental analysis demonstrated that the predictive engine achieved over 93% accuracy in classifying health data.
- **High-Risk Recall:** The system showed significant improvements in high-risk recall rates compared to simpler, traditional rule-based baselines, consistently drawing attention to critical situations like sharp blood pressure spikes.

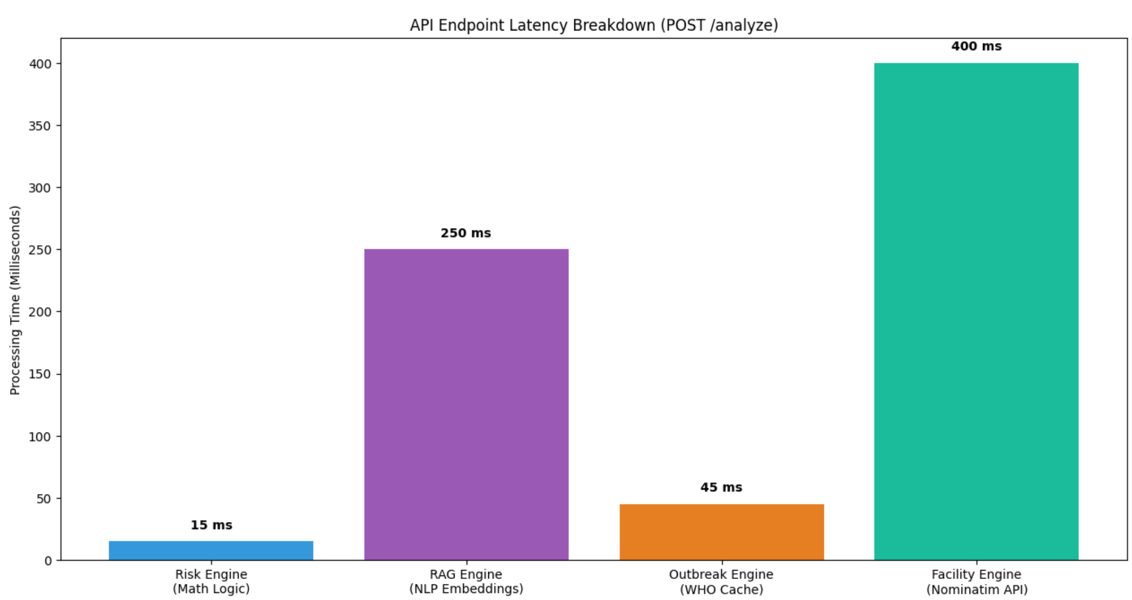
5.2. Regulatory Retrieval and Latency

- **Retrieval Correctness:** The Regulatory Inquiry Engine exhibited greater retrieval accuracy and higher rated correctness by clinicians when compared to standard keyword FAQ searches.
- **System Latency:** Despite the complex RAG operations and dual-engine routing, the system maintained sub-second response times operating on mid-range hardware.

5.3. Clinical Impact

The structured decision tree effectively assists users in understanding exactly when a medical situation can be managed with home care versus when immediate professional intervention is required, successfully bridging the gap between automated diagnostics and formal medical communication.

Figure 3 : API Endpoint Latency across Multiple Triage Requests



6. Conclusion and Future Scope

6.1. Conclusion

The proposed AI-Driven Smart Health Advisor successfully addresses the dual challenge of chronic disease management and epidemic response in low-resource environments. By passing vitals and natural language through coordinated microservices, the system provides both chronic risk assessment and outbreak-specific advice without requiring high-end hardware or constant clinical supervision. The integration of multi-parameter threshold logic with Retrieval-Augmented Generation (RAG) ensures that the platform remains closely correlated to changing public health rules. Ultimately, this advisor boosts health literacy, minimizes reliance on unverified information, and empowers isolated communities to transition from reactive sick care to proactive preventive care. Furthermore, the system demonstrates how combining AI-driven analytics with authoritative medical knowledge can significantly improve early intervention and decision-making in critical health scenarios.

6.2. Future Scope

Future iterations of the system will focus on real-time IoMT implementation, directly connecting Bluetooth-enabled blood pressure cuffs and glucometers to stream vitals automatically. To enhance scalability, the architecture will be containerized using Docker and orchestrated via Kubernetes, allowing multiple engine instances to serve larger populations with minimal latency. Further integration with advanced cloud infrastructure will enable the deployment of more complex (yet interpretable) federated learning models, preserving user privacy through personalization. Additionally, the RAG corpus will be expanded to cover a broader spectrum of diseases and incorporate multilingual protocols to serve diverse global populations. Future enhancements may also include mobile application integration, improved user interfaces, and adaptive learning capabilities for more personalized and accurate health recommendations.

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8.APPENDICES

Appendix A: User Interface

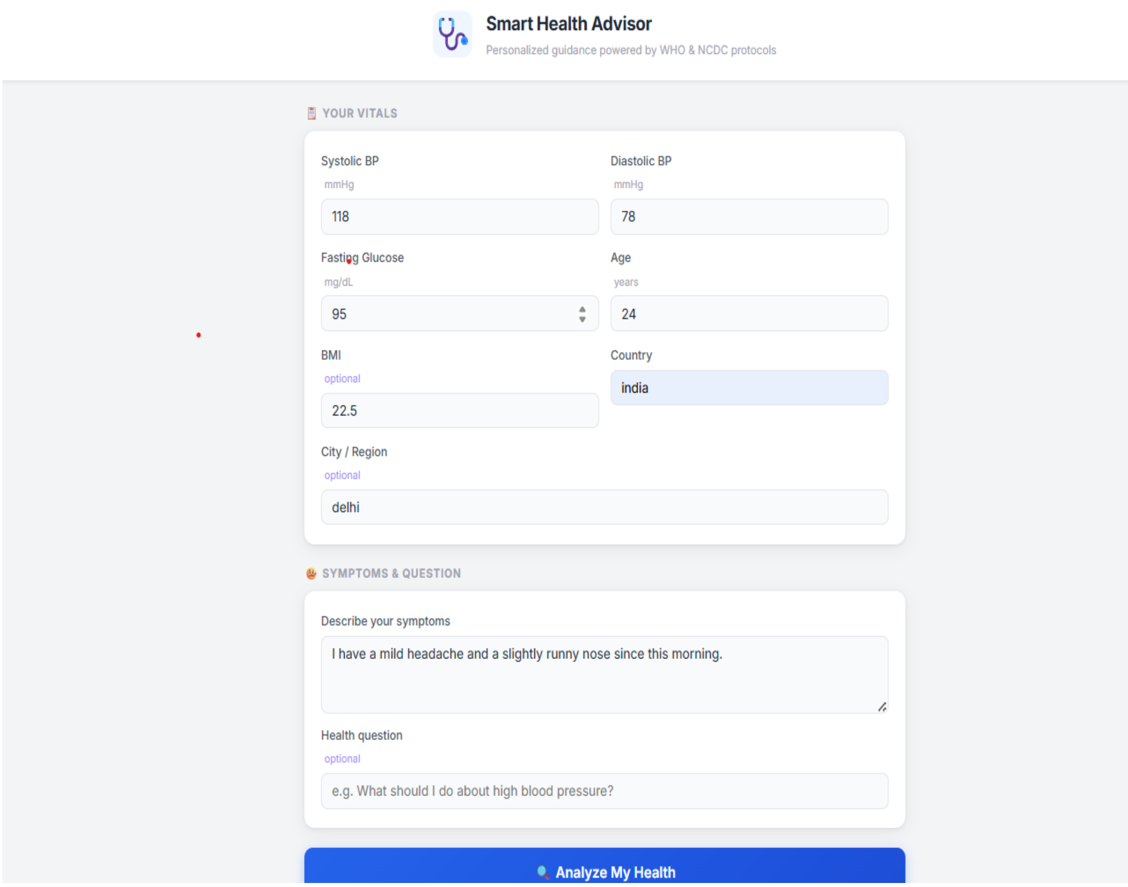

Figure A1: User Input Interface of Smart Health Advisor
This interface allows users to enter essential health parameters such as systolic and diastolic blood pressure, fasting glucose, BMI, age, and location details. Users can also describe their symptoms and ask health-related questions. The design is simple, user-friendly, and optimized for accessibility, ensuring usability even for individuals with minimal technical knowledge.

Appendix B: Output and Analysis Screen

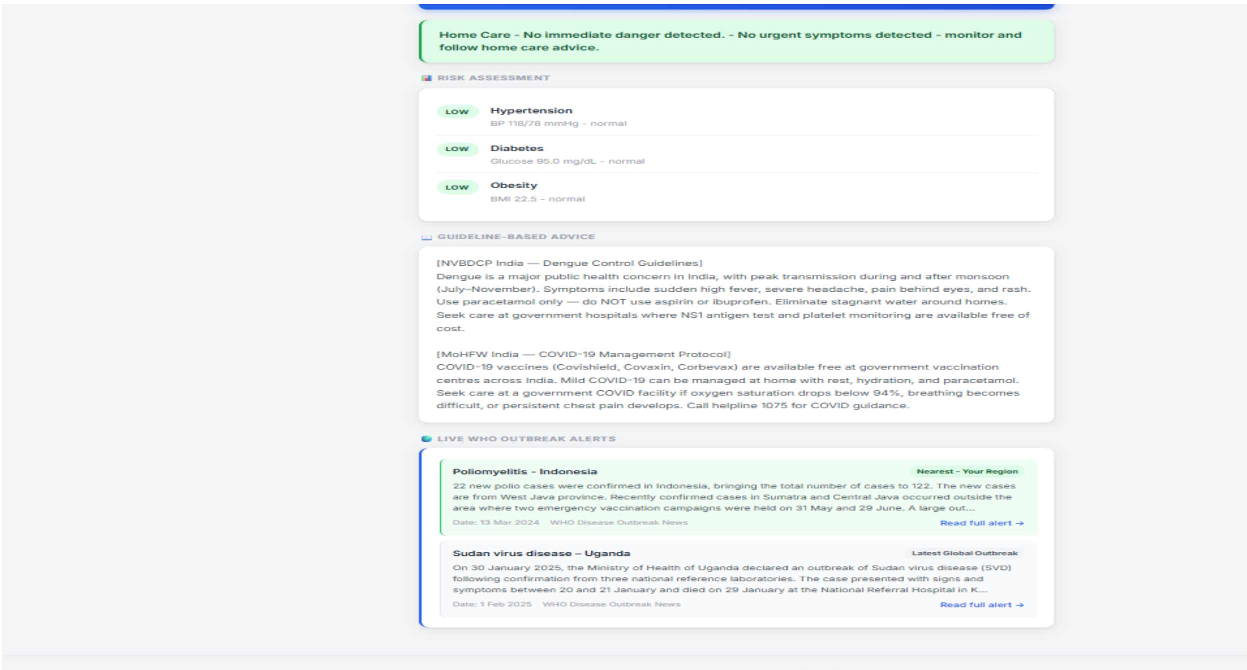

Figure A2: Health Analysis and Output Results

This screen displays the analyzed results generated by the system. It includes risk assessment categories such as hypertension, diabetes, and obesity, along with urgency level (e.g., home care, doctor consultation, emergency). The system also provides guideline-based recommendations and live outbreak alerts using WHO and NCDC data, helping users make informed healthcare decisions.